

Smart Recommendation Engine Module

The Smart Recommendation Engine Module is the core intelligence of the Smart Car Recommendation System. It is responsible for analyzing user inputs and generating personalized car suggestions based on their needs and preferences. This module takes inputs such as budget, fuel type, usage (city driving, long drives, family use), and priorities (mileage, performance, safety). Based on these inputs, the system processes data using predefined logic and algorithms to suggest the top 3–5 most suitable cars. The recommendation engine does not simply list cars but ranks them based on relevance. It evaluates multiple parameters such as price compatibility, fuel efficiency, maintenance cost, and feature match to ensure accurate suggestions. Additionally, the module provides reasoning for each recommendation, helping users understand why a particular car is suitable for them. For example, it may highlight “best mileage in budget” or “ideal for long-distance travel.” The engine continuously interacts with the car database and filtering system to fetch real-time data and improve recommendation accuracy. It can also be enhanced in the future using Machine Learning models to make smarter and more dynamic predictions.

Technologies Used: - HTML: For input forms where users provide their requirements. - CSS: For designing user-friendly input interfaces. - JavaScript: For handling dynamic inputs and sending data to the backend. - Python (Flask): For implementing recommendation logic, API handling, and data processing. - Database (MongoDB/MySQL): For retrieving car data used in generating recommendations. Overall, the Smart Recommendation Engine Module provides intelligent, personalized, and efficient car suggestions, making the decision-making process faster and easier for users.